RAYCASTED: A RAYCAST END-TO-END DETECTION MODEL FOR LIGHTWEIGHT AND BOUNDARY-AWARE NUCLEI DETECTION IN HISTOPATHOLOGY IMAGES

SKRIPSI

THE SUSTAINABLE DEVELOPMENT GOALS
Industry, Innovation and Infrastructure
Affordable and Clean Energy
Climate Action

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ENDORSEMENT PAGE

RAYCASTED: A RAYCAST END-TO-END DETECTION MODEL FOR LIGHTWEIGHT AND BOUNDARY-AWARE NUCLEI DETECTION IN HISTOPATHOLOGY IMAGES

THESIS

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PREFACE

Contoh: Puji syukur ke hadirat Allah SWT atas limpahan rahmat, karunia, serta petunjuk-Nya sehingga tugas akhir berupa penyusunan skripsi ini telah terselesaikan dengan baik. Dalam hal penyusunan tugas akhir ini penulis telah banyak mendapatkan arahan, bantuan, serta dukungan dari berbagai pihak. Oleh karena itu pada kesempatan ini penulis mengucapkan terima kasih kepada:

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Akhir kata penulis berharap semoga skripsi ini dapat memberikan manfaat bagi kita semua, aamiin.

Catatan: setiap nama yang dituliskan boleh disertai dengan alasan berterima kasih.

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NOMENCLATURE AND ABBREVIATION

[SAMPLE]

b	=	bias
$K(x_i, x_j)$	=	fungsi kernel
y	=	kelas keluaran
C	=	parameter untuk mengendalaikan besarnya pertukaran antara penalti variabel slack dengan ukuran margin
L_D	=	persamaan Lagrange dual
L_P	=	persamaan Lagrange primal
$\mathbf{w}$	=	vektor bobot
$\mathbf{x}$	=	vektor masukan
ANFIS	=	Adaptive Network Fuzzy Inference System
ANSI	=	American National Standards Institute
DAG	=	Directed Acyclic Graph
DDAG	=	Decision Directed Acyclic Graph
HIS	=	Hue Saturation Intensity
QP	=	Quadratic Programming
RBF	=	Radial Basis Function
RGB	=	Red Green Blue
SV	=	Support Vector
SVM	=	Support Vector Machines

INTISARI

Intisari ditulis menggunakan bahasa Indonesia dengan jarak antar baris 1 spasi dan maksimal 1 halaman. Intisari sekurang-kurangnya berisi tentang latar belakang dan tujuan penelitian, metodologi yang digunakan, hasil penelitian, kesimpulan dan implikasi, dan Kata kunci yang berhubungan dengan penelitian.

Kata Kunci ditulis maksimal 5 kata yang paling berhubungan dengan isi skripsi. Silakan mengacu pada ACM / IEEE *Computing classification* jika Anda adalah mahasiswa Sarjana TI http://www.acm.org/about/class/ atau mengacu kepada IEEE keywords http://www.ieee.org/documents/taxonomy_v101.pdf jika Anda berasal dari Prodi Sarjana TE.

Kata kunci : Kata kunci 1, Kata kunci 2, Kata kunci 3, Kata kunci 4, Kata kunci 5

Contoh Abstrak Teknik Elektro:

"Penelitian ini bertujuan untuk mengembangkan sistem pengendalian suhu ruangan dengan menggunakan microcontroller. Metodologi yang digunakan adalah desain sirkuit, implementasi sistem pengendalian, dan pengujian performa. Hasil penelitian menunjukkan bahwa sistem pengendalian suhu ruangan yang dikembangkan mampu mengendalikan suhu ruangan dengan akurasi sebesar ±0,5°C. Kesimpulan dari penelitian ini adalah sistem pengendalian suhu ruangan yang dikembangkan efektif dan efisien.

Kata kunci: microcontroller, sistem pengendalian suhu, akurasi."

Contoh Abstrak Teknik Biomedis:

"Penelitian ini bertujuan untuk mengevaluasi keefektifan prototipe alat pemantau denyut jantung berbasis elektrokardiogram (ECG) untuk pasien jantung. Metodologi yang digunakan meliputi desain dan pembuatan prototipe, pengujian dengan pasien, dan analisis data. Hasil penelitian menunjukkan bahwa prototipe alat pemantau denyut jantung berbasis ECG memiliki akurasi yang baik dan mampu memantau denyut jantung pasien secara efektif. Kesimpulan dari penelitian ini adalah prototipe alat pemantau denyut jantung berbasis ECG merupakan solusi yang efektif dan efisien untuk memantau pasien jantung.

Kata kunci: elektrokardiogram, alat pemantau denyut jantung, akurasi."

Contoh Abstrak Teknologi Informasi:

"Penelitian ini bertujuan untuk mengevaluasi keamanan dan privasi pengguna aplikasi media sosial terpopuler. Metodologi yang digunakan meliputi analisis kebijakan privasi dan pengaturan keamanan, pengujian penetrasi, dan survei pengguna. Hasil penelitian menunjukkan bahwa beberapa aplikasi media sosial memiliki kebijakan privasi yang kurang jelas dan rendahnya tingkat keamanan. Kesimpulan dari penelitian ini adalah pentingnya meningkatkan kebijakan privasi dan tingkat keamanan pada aplikasi media sosial untuk melindungi privasi dan data pengguna.

Kata kunci: media sosial, keamanan, privasi, pengguna."

ABSTRACT

Abstract ditulis italic (miring) menggunakan bahasa Inggris dengan jarak antar baris 1 spasi dan maksimal 1 halaman. Abstract adalah versi Bahasa Inggris dari intisari. Abstract dapat ditulis dalam beberapa paragraf. Baris pertama paragraph harus menjorok ke dalam sekitar 1 cm. Tidak dsarankan menggunakan mesin penerjemah melainkan tulis ulang.

Keywords : Keyword 1, Keyword 2, Keyword 3, Keyword 4, Keyword 5

CHAPTER I
INTRODUCTION

1.1 Background

Tumor-infiltrating lymphocytes (TILs) density has emerged as an established prognostic biomarker in multiple cancer types. The International TILs Working Group has developed standardized guidelines for TILs assessment, which recommend estimating the percentage of stromal area occupied by lymphocytes under a microscope [1]. This scoring methodology is fundamentally an *area-based* measurement, which depends on accurately delineating cell boundaries rather than merely locating their centroids.

However, manual TILs assessment presents significant challenges. The process is time-consuming, suffers from substantial inter-observer variability among pathologists, and becomes impractical when applied to whole-slide images (WSIs) at clinical scale. These limitations have motivated the development of automated approaches for nuclei detection and segmentation in histopathology images.

Critically, practical deployment in clinical settings presents additional constraints: many healthcare facilities, particularly in resource-constrained environments, require models that can run efficiently on edge devices rather than demanding powerful centralized computing infrastructure. This deployment constraint creates a fundamental tension: models must be accurate enough for clinical decision-making yet lightweight enough for practical use in real-world clinical workflows.

Automated approaches for nuclei detection in histopathology images can be broadly categorized into three paradigms, each with distinct trade-offs. **Instance segmentation methods** such as U-Net [2] and Mask R-CNN [3] produce accurate boundary delineations but are computationally expensive, particularly for whole-slide image analysis. **Vision transformer-based models** like CellViT [4] demonstrate strong performance on histopathology tasks, but their self-attention mechanisms impose quadratic complexity in both memory and computational requirements, making them less suitable for resource-constrained deployment. In contrast, **object detection frameworks** including Faster R-CNN [5] and YOLO [6] offer fast, lightweight inference at the cost of discarding boundary geometry information through bounding box representations, which is insufficient for area-based TILs scoring.

A fundamental challenge across all automated approaches is the **clumped cell problem**. Densely clustered nuclei are nearly impossible to separate in segmentation masks because fully convolutional network (FCN) based models produce contiguous predictions that merge adjacent cells, leading to systematic undercounting. The computational

burden of FCN-based instance segmentation stems not from the convolutional architecture itself, but from the costly post-processing required to separate individual instances. For example, HoVer-Net [7] addresses this challenge using horizontal and vertical distance maps, while StarDist [8] represents cells as star-convex polygons to preserve boundary shape while maintaining per-cell separation. However, these methods still rely on two-stage pipelines (dense prediction followed by post-processing) and typically use non-maximum suppression (NMS), which suppresses overlapping predictions and risks undercounting in precisely the clumped regions where accurate counting matters most.

Recent advances have sought to address the NMS dependency through end-to-end detection frameworks. LSP-DETR [9] adopts a DETR-style framework with learned assignment mechanisms, eliminating the need for NMS while representing nuclei as star-convex polygons. However, its transformer-based architecture remains computationally demanding for edge deployment scenarios. Meanwhile, CellPose-SAM [10] has demonstrated superhuman generalization for cellular segmentation, though its computational profile may not be optimized for real-time edge deployment in clinical workflows.

Research Gap: No existing method combines polygon-based cell representation with a single-stage, end-to-end FCN detection architecture that eliminates NMS dependency while maintaining the lightweight computational profile necessary for edge deployment. This thesis addresses this gap by proposing **RayCastED**, an FCN-based adaptation of LSP-DETR that re-frames the star-convex representation as a *raycast* representation (centroid plus fixed-angle radial rays), integrated directly into a YOLO-based detection head. By replacing the transformer backbone with an FCN, RayCastED maintains the lightweight profile needed for edge deployment through TensorRT and ONNX Runtime while preserving the end-to-end assigner-based framework that eliminates NMS, thereby addressing both boundary accuracy and clumped-cell separation with detection-level speed.

Key Points Outline

The following outline summarizes the main points covered in this section:

1. **Clinical significance of TILs.** Tumor-infiltrating lymphocytes (TILs) density is an established prognostic biomarker in several cancers (e.g., the International TILs Working Group guidelines for breast cancer).
2. **How TILs are scored.** Pathologists estimate the percentage of stromal area occupied by lymphocytes under a microscope. This is fundamentally an *area-based* measurement, which depends on accurately delineating cell boundaries, not just locating them.
3. **Manual problem.** This scoring is time-consuming, suffers from inter-observer vari-

ability, and is impractical at whole-slide image (WSI) scale. Furthermore, practical deployment in clinical settings, especially in resource-constrained environments, benefits from models that can run efficiently on edge devices.

4. **Automated approaches and trade-offs.**
   - Instance segmentation (U-Net, Mask R-CNN): accurate boundaries but computationally expensive
   - Transformer-based models (CellViT, DINO-based): strong performance but self-attention imposes quadratic complexity in memory and compute. FCN-based architectures will almost always be lighter and more efficient due to their fixed, local receptive fields, making them inherently better suited for resource-constrained edge deployment.
   - Object detection (YOLO, Faster R-CNN): fast and lightweight but bounding boxes discard boundary geometry, insufficient for area-based scoring.
5. **The clumped cell problem.** A fundamental challenge across all approaches: densely clustered nuclei are nearly impossible to separate in segmentation masks because FCN-based models produce contiguous predictions that merge adjacent cells, leading to systematic undercounting. Note that the computational burden of FCN-based instance segmentation stems not from the convolutional architecture itself, but from the costly post-processing required to separate individual instances, which will be discussed in detail in later sections.
6. **Polygon-based detection as mitigation.** StarDist and similar methods represent cells as star-convex polygons, preserving boundary shape while maintaining per-cell separation. However, these rely on a two-stage pipeline (dense prediction $\rightarrow$ post-processing) and still use NMS, which suppresses overlapping predictions and risks undercounting in precisely the clumped regions that matter most. LSP-DETR addresses the NMS issue by adopting an end-to-end DETR-style framework with learned assignment, but its transformer-based architecture is computationally demanding for edge deployment.
7. **Gap and contribution.** No existing method combines polygon-based cell representation with a single-stage, end-to-end FCN detection architecture that avoids NMS dependency. This thesis proposes **RayCastED**, an FCN-based adaptation of LSP-DETR that re-frames the star-convex representation as a *raycast* representation (centroid + fixed-angle radial rays), integrated directly into a YOLO-based detection head. By replacing the transformer backbone with an FCN, RayCastED maintains the lightweight profile needed for edge deployment (TensorRT, ONNX Runtime) while preserving the end-to-end assigner-based framework that eliminates NMS, addressing both boundary accuracy and clumped-cell separation with detection-level

speed.

1.2 Problem Statement

Based on the background discussed above, the following problems are identified:

1. **Accuracy-deployment trade-off:** Existing methods force a choice between clinical accuracy and practical deployability. Segmentation models provide accurate boundaries but require computational resources unavailable in many clinical settings, while detection models offer lightweight deployment but lack the boundary geometry essential for TILs scoring.
2. **Clumped cell undercounting:** FCN-based instance segmentation models produce contiguous predictions for densely clustered nuclei, merging adjacent cells and leading to systematic undercounting that compromises both cell counting accuracy and area measurement precision.
3. **Computational inefficiency:** The computational burden of instance segmentation stems largely from costly post-processing required to separate individual instances, rather than the convolutional architecture itself, making real-time analysis impractical in clinical workflows.
4. **Loss of boundary geometry:** Object detection models disregard cell boundary geometry by outputting only bounding boxes, which is insufficient for area-based TILs scoring that requires accurate cell delineation for clinical decision-making.
5. **NMS-based suppression errors:** Methods relying on non-maximum suppression risk undercounting in densely clumped regions by suppressing overlapping predictions, precisely where accurate counting matters most for clinical assessment.
6. **Deployment inaccessibility:** Vision Transformer-based models achieve better cell separation but impose quadratic computational complexity, making them impractical for resource-constrained edge deployment and limiting accessibility to well-funded institutions only.

1.3 Objectives

To address the identified problems, this thesis has the following objectives:

1. Develop **RayCastED**, a nuclei detection model that:
   - preserves cell boundary geometry through a raycast polygon representation (centroid + fixed-angle radial rays), addressing the lack of geometric information in conventional object detection models (Problem 3);
   - eliminates costly instance separation post-processing by predicting per-cell poly-

gons directly from a detection head, addressing both the clumped cell separation issue (Problem 1) and the post-processing overhead (Problem 2);

- replaces NMS with a learned end-to-end assignment mechanism, preventing under-counting in densely clumped regions (Problem 4); and
- operates on an FCN backbone to maintain a lightweight computational profile suitable for edge deployment via TensorRT and ONNX Runtime (Problem 5).

2. Evaluate the performance of RayCastED by comparing it against prior nuclei detection and segmentation models on standard benchmarks.
3. Evaluate the performance of RayCastED on edge devices.

1.4 Research Scope

This research is bounded by the following scope limitations:

1. **Data characteristics:** The study uses only H&E stained histopathology images with approximately 0.25 MPP (microns per pixel) resolution from publicly available secondary datasets. The datasets encompass multiple tissue types/organs.
2. **Computational constraints:** Model evaluation is performed on RTX 5070ti hardware. Baseline comparisons are limited to models compatible with this hardware configuration.
3. **Validation approach:** The research employs computational validation using standard metrics. Pathologist or expert validation is not included in this study.
4. **Technical boundaries:** There are no limitations on nuclei types, sizes, or detection scope. No strict constraints are imposed on input patch sizes or memory usage during model training and inference.

1.5 Research Novelty

The novelty of this research lies in the development of RayCastED, which represents the first architecture that combines three key elements: (1) star-convex polygon representation for preserving cell boundary geometry, (2) FCN backbone for maintaining lightweight computational profile, and (3) end-to-end learned assignment mechanism that eliminates NMS dependency. This unique combination addresses the fundamental trade-off between boundary accuracy and computational efficiency in nuclei detection for histopathology images.

1.6 Research Contributions

The contributions of this research are as follows:

1. **RayCastED model implementation:** A complete implementation of the proposed polygon-based detection model with raycast representation, integrated into a YOLO-based detection head with end-to-end assignment mechanism.
2. **Performance benchmarks on standard datasets:** Comprehensive evaluation of RayCastED on publicly available histopathology datasets, comparing performance against baseline models including instance segmentation, object detection, and transformer-based approaches.
3. **Edge deployment validation:** Validation of RayCastED's deployment capabilities on edge devices through TensorRT and ONNX Runtime conversion, demonstrating practical feasibility for resource-constrained clinical environments.
4. **Comparison against baseline models:** Systematic comparison with prior nuclei detection and segmentation methods to quantify improvements in accuracy, computational efficiency, and clumped-cell separation performance.
5. **Reproducible codebase:** A publicly available, well-documented implementation that enables reproducibility and provides a foundation for future research in polygon-based detection for computational pathology.

1.7 Presentation Structure

Sistematika penulisan berisi pembahasan apa yang akan ditulis di setiap bab. Sistematika pada umumnya berupa paragraf yang setiap paragraf mencerminkan bahasan setiap Bab. Bagian ini akan ditulis setelah seluruh bab dalam skripsi ini selesai disusun.

CHAPTER II
TINJAUAN PUSTAKA DAN DASAR TEORI

2.1 Literature Review

This section reviews prior work on nuclei detection and segmentation in histopathology images, focusing on methods most relevant to the proposed RayCastED model. The reviewed works are organized chronologically, tracing the evolution from polygon-based segmentation to end-to-end detection frameworks.

StarDist

StarDist [8] introduced the star-convex polygon representation for cell detection, originally developed for fluorescence microscopy. In their extended work [11], Weigert and Schmidt adapted StarDist to histopathology images, demonstrating its applicability to the CoNIC challenge where it achieved first place in the segmentation and classification task. StarDist predicts a set of radial distances from each cell centroid to its boundary at fixed angular intervals, producing a star-convex polygon that preserves boundary geometry. However, StarDist relies on a two-stage pipeline: it first generates a dense probability map and then applies non-maximum suppression (NMS) to filter overlapping predictions. This NMS step risks undercounting in densely clumped regions where overlapping predictions are suppressed, and the dense prediction approach introduces computational overhead from post-processing.

CPP-Net

CPP-Net [12] addressed a key limitation of earlier polygon-based methods by improving how contextual information is used during polygon prediction. Rather than predicting the centroid-to-boundary distances from a single centroid pixel, CPP-Net samples a point set within each cell and fuses their predictions using a Confidence-based Weighting Module. The authors also introduced a Shape-Aware Perceptual (SAP) loss that constrains predicted polygon shapes. CPP-Net achieved state-of-the-art results on DSB2018, BBBC06, and PanNuke datasets. Despite its improvements in prediction robustness, CPP-Net still operates within a dense prediction framework with NMS-based post-processing, inheriting the same limitations as StarDist regarding clumped cell separation and computational overhead.

LKCell

LKCell [13] proposed an efficient approach to nuclei instance segmentation by leveraging large convolution kernels to achieve large receptive fields with reduced com-

putational cost. The method transfers pre-trained large convolution kernel models to the medical domain for the first time and designs a new segmentation decoder based on these kernels. LKCell achieved state-of-the-art performance (0.5080 mPQ) on PanNuke with only 21.6% of the FLOPs compared to the previous leading method, demonstrating that FCN-based architectures can be both accurate and computationally efficient. However, LKCell follows a traditional dense prediction paradigm and does not address the NMS dependency or the clumped cell separation problem, as it focuses primarily on computational efficiency rather than the detection-accuracy trade-off.

CellPose-SAM

CellPose-SAM [14] combined the CellPose framework with the Segment Anything Model (SAM) to achieve what the authors describe as superhuman generalization for cellular segmentation. By leveraging SAM's powerful visual features, CellPose-SAM demonstrates strong cross-domain performance without requiring extensive domain-specific training data. While its generalization capability is impressive, CellPose-SAM is a generalist segmentation model rather than a lightweight detection framework, and its computational profile makes it less suitable for real-time edge deployment in resource-constrained clinical settings.

LSP-DETR

LSP-DETR [9] represents the most direct precursor to this work. It introduced a fully end-to-end framework for nuclei segmentation that represents nuclei as star-convex polygons within a DETR-style transformer architecture. By using a learned assignment mechanism instead of NMS, LSP-DETR eliminates the undercounting problem in clumped regions. The authors also proposed a novel radial distance loss that allows overlapping nuclei segmentation to emerge naturally. LSP-DETR demonstrated strong generalization across tissues and was over five times faster than the next-fastest leading method. However, its transformer-based backbone imposes quadratic complexity, making it computationally demanding for edge deployment. RayCastED builds directly on LSP-DETR's polygon representation and end-to-end assignment mechanism while replacing the transformer backbone with an FCN to achieve a lightweight computational profile suitable for resource-constrained environments.

Summary and Research Opportunity

The reviewed methods reveal a clear trade-off landscape: polygon-based methods (StarDist, CPP-Net) preserve boundary geometry but depend on NMS; efficient FCN methods (LKCell) reduce computation but do not address NMS or boundary geometry; generalist models (CellPose-SAM) generalize well but are computationally heavy; and

end-to-end transformer methods (LSP-DETR) eliminate NMS but rely on expensive transformer backbones. No existing method combines polygon-based cell representation, NMS-free end-to-end detection, and lightweight FCN-based computation. RayCastED addresses this gap by adapting LSP-DETR's polygon and assignment framework to an FCN backbone, targeting deployment on resource-constrained edge hardware.

2.2 Theoretical Basis

2.2.1 Histopathological Foundations

Histopathology is the study of tissue disease through microscopic examination of tissue samples. In clinical practice, tissue specimens are collected through biopsy or surgical resection and processed for microscopic analysis. The tissue undergoes fixation (typically in formalin) to preserve cellular structures, followed by embedding in paraffin wax, sectioning into thin slices, and mounting on glass slides for staining and visualization.

The cell nucleus is a membrane-bound organelle containing genetic material and serves as the primary structure of interest in histopathological analysis. Nuclear morphology, including size, shape, texture, and spatial distribution, provides critical diagnostic information. In cancer diagnosis, malignant cells often exhibit nuclear atypia: enlarged nuclei, irregular shapes, hyperchromasia (increased staining intensity), and abnormal mitotic figures.

A fundamental challenge in nuclei analysis is that nuclei in tissue sections are densely packed and frequently overlap, forming clumped clusters where individual boundaries are difficult to distinguish even for expert pathologists. This overlapping occurs because tissue sections are typically cut at a thickness of 3–5 μm, which is comparable to or smaller than the diameter of many nuclei (typically 5–10 μm). As a result, multiple nuclei at different depths within the section are projected onto the same 2D image plane, creating the appearance of clumped or touching nuclei. This is not merely an artifact of imaging resolution but an inherent property of projecting 3D tissue architecture onto a 2D plane, motivating the development of automated methods capable of both accurately delineating individual nuclear boundaries and correctly separating touching or overlapping nuclei.

2.2.1.1 Tumor-Infiltrating Lymphocytes and Clinical Significance

Tumor-infiltrating lymphocytes (TILs) are immune cells, primarily lymphocytes and plasma cells, that have migrated into the tumor microenvironment. The presence and density of TILs reflect the host immune response against the tumor and have been established as a prognostic biomarker in multiple cancer types. Higher TILs levels are generally associated with improved outcomes, particularly in triple-negative and HER2-positive breast cancers, where they predict better response to chemotherapy and longer

survival.

The International TILs Working Group, convened by Salgado et al. [1], established standardized guidelines for TILs assessment in clinical practice. The recommended metric is **stromal TILs (sTILs)**, defined as the proportion of stromal area occupied by mononuclear inflammatory cells within the invasive tumor border:

$$\text{sTILs (\%)} = \frac{\text{Area of mononuclear inflammatory cells in stroma}}{\text{Total intratumoral stromal area}} \times 100 \tag{2-1}$$

Only lymphocytes and plasma cells are included; polymorphonuclear leukocytes are excluded. Areas of necrosis, crush artifact, fibrosis, and hyalinization at the tumor border are also excluded from the assessment. The score is reported as a single average percentage per tumor on H&E-stained full-tissue sections.

Clinically, tumors with sTILs $\geq$ 50% are classified as lymphocyte-predominant breast cancer (LPBC), indicating a strong immune response. Additional commonly used thresholds include sTILs < 10% (TILs-low) and sTILs $\geq$ 30% (TILs-high in many triple-negative breast cancer studies).

This scoring methodology is fundamentally an *area-based* measurement: it requires accurately delineating the boundary of each lymphocyte to compute its area contribution, not merely detecting its presence or counting its centroid. This distinction has direct implications for the design of automated nuclei analysis systems. Object detection models that output only bounding boxes or centroids are insufficient for sTILs scoring because they discard boundary geometry. Accurate sTILs computation requires methods that can both detect individual cells and delineate their boundaries, particularly in densely clumped regions where cell boundaries are ambiguous.

2.2.2 Digital Pathology Imaging

2.2.2.1 Tissue Slicing and Scanning

The digitization of histopathology slides produces whole-slide images (WSIs) through specialized scanners that capture high-resolution images of the entire tissue section. Modern slide scanners, such as those from Hamamatsu, Leica, and 3DHistech, use line-sensor or area-sensor cameras mounted on motorized stages to systematically capture overlapping tiles that are subsequently stitched into a single large image.

WSIs are typically stored in multi-resolution pyramidal formats (e.g., Aperio SVS, Hamamatsu NDP, or generic TIFF), where each level provides a different magnification. The spatial resolution of a WSI is specified in microns per pixel (MPP), which depends on the objective lens magnification used during scanning. Common resolutions range

from approximately 0.25 MPP (equivalent to 40× magnification) to 0.5 MPP (20×). At 0.25 MPP, a typical biopsy slide can produce an image exceeding 50,000 × 50,000 pixels, necessitating patch-based processing strategies for computational analysis.

2.2.2.2 Staining and its Variants

Tissue staining is essential because most biological tissues are nearly transparent under standard brightfield microscopy. Stains introduce contrast by selectively binding to tissue components through chemical interactions. The most widely used staining protocol in histopathology is Hematoxylin and Eosin (H&E) staining.

Hematoxylin is a basic (positively charged) dye that binds to acidic structures such as DNA and RNA, coloring nuclei dark blue or purple. Eosin is an acidic (negatively charged) dye that binds to basic structures such as cytoplasmic proteins and extracellular matrix, coloring them pink. The combination produces the characteristic appearance of H&E-stained tissue: blue-purple nuclei against a pink cytoplasmic and stromal background.

Beyond H&E, several specialized stains are used in clinical practice: Periodic Acid-Schiff (PAS) for basement membranes and glycogen, Masson's Trichrome for collagen fibers, and immunohistochemical (IHC) stains for specific protein markers. Each stain produces distinct color characteristics that can be mathematically decomposed for computational analysis.

2.2.2.3 Beer-Lambert Law and Optical Density

The interaction of light with stained tissue follows the Beer-Lambert law, which describes the relationship between light absorption and the concentration of absorbing material. For a stained tissue sample, the transmitted intensity I at a given wavelength is related to the incident intensity I_0 by:

$$I = I_0 \cdot e^{-\alpha \cdot c \cdot d} \tag{2-2}$$

where α is the molar absorptivity (extinction coefficient), c is the concentration of the stain, and d is the path length through the tissue. The optical density (OD) is defined as:

$$\text{OD} = -\log_{10}\left(\frac{I}{I_0}\right) = \alpha \cdot c \cdot d \cdot \log_{10}(e) \tag{2-3}$$

In the context of digital pathology, the RGB pixel values of a scanned image are proportional to transmitted light intensity. Converting from RGB to optical density space is essential because OD is linearly additive: if multiple stains are present, the total OD

at each pixel equals the sum of the OD contributions from each individual stain. This linearity property underpins color deconvolution methods.

2.2.2.4 Stain Vectors

A stain vector represents the characteristic color signature of a specific stain in optical density space. For an RGB image, each stain is described by a 3-element unit vector $\mathbf{s} = [s_R, s_G, s_B]^T$ in OD space, where each component indicates the stain's relative absorption at the red, green, and blue channels.

For H&E staining, the two stain vectors are:

- **Hematoxylin vector** $\mathbf{s}_H$: characterizes the absorption pattern of hematoxylin, which strongly absorbs in the blue-green range, producing its characteristic dark blue appearance.
- **Eosin vector** $\mathbf{s}_E$: characterizes the absorption pattern of eosin, which absorbs primarily in the green range, producing its pink appearance.

When two stains are present, the stain matrix S is formed by concatenating the stain vectors as columns: $S = [\mathbf{s}_H \quad \mathbf{s}_E]$. For three-stain protocols, a third column is added.

2.2.2.5 Stain Vector Estimation Methods

Since stain appearance varies across scanners, laboratories, and tissue types, the stain vectors must be estimated from each individual image or dataset. Several methods exist for automatic stain vector estimation:

Macenko's method estimates stain vectors by analyzing the distribution of pixel OD values in the image. Pixels are converted to OD space, and those with extreme OD values (typically the top 1%) are identified. Singular Value Decomposition (SVD) is then applied to these pixels, and the two principal angles corresponding to the strongest eigenvectors define the stain directions. The resulting vectors are then projected onto a predefined stain color space to assign hematoxylin and eosin identities.

Vahadane's method uses non-negative matrix factorization (NMF) to simultaneously estimate the stain matrix and the stain density maps. This approach enforces sparsity constraints and non-negativity, which are physically meaningful since stain concentrations cannot be negative.

The choice of estimation method affects downstream stain normalization quality. Macenko's method is widely used due to its simplicity and robustness, while Vahadane's method produces more accurate separations at the cost of higher computational complexity.

2.2.2.6 Color Deconvolution

Color deconvolution is the process of separating an H&E-stained image into its individual stain channels. Given the stain matrix S and the OD image $\mathbf{v}$ (a pixel vector in OD space), the stain concentrations $\mathbf{c}$ for each pixel are obtained by solving the linear system:

$$\mathbf{v} = S \cdot \mathbf{c} \tag{2-4}$$

Since S is a 3×2 matrix (for H&E) or 3×3 (for three stains), the system can be solved using the pseudoinverse:

$$\mathbf{c} = S^{-1} \cdot \mathbf{v} \tag{2-5}$$

where S^{-1} denotes the pseudoinverse (or inverse, for square S). The resulting stain concentration maps $\mathbf{c}$ provide separate grayscale images representing the hematoxylin and eosin density at each pixel. The hematoxylin channel is particularly useful for nuclei detection algorithms, as it isolates nuclear staining from cytoplasmic and stromal background.

2.2.2.7 Stain Normalization

Stain normalization addresses the significant color variation that exists across WSIs captured from different scanners, hospitals, or staining batches. This variation, if uncorrected, degrades the performance of deep learning models trained on data from one source and applied to another.

The standard approach to stain normalization involves three steps: (1) estimate the stain matrix S_{source} from the source image using one of the methods described above, (2) compute the stain concentration map $\mathbf{c} = S_{\text{source}}^{-1} \cdot \mathbf{v}_{\text{source}}$, and (3) reconstruct the normalized image using a target stain matrix S_{target}: $\mathbf{v}_{\text{normalized}} = S_{\text{target}} \cdot \mathbf{c}$, then convert back from OD to RGB space.

This process preserves the structural content of the image while standardizing color appearance. Stain normalization is commonly applied as a preprocessing step before training or inference in computational pathology pipelines.

2.2.3 Convolutional Neural Networks

2.2.3.1 Convolution and Feature Extraction

Convolutional Neural Networks (CNNs) form the backbone of modern computer vision. The fundamental operation in a CNN is the convolution, where a learnable filter

$\mathbf{W}$ of size $k \times k$ is slid across the input feature map $\mathbf{X}$ to produce an output feature map $\mathbf{Y}$. For a spatial position (i, j), the convolution output is computed as:

$$\mathbf{Y}(i,j) = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} \mathbf{W}(m,n) \cdot \mathbf{X}(i+m, j+n) + b \tag{2-6}$$

where b is a learnable bias term. Each convolutional layer applies multiple such filters, producing a set of feature maps that capture local spatial patterns such as edges, textures, and more complex structures. By stacking convolutional layers with nonlinear activation functions (typically ReLU) and pooling layers that reduce spatial dimensions, CNNs learn hierarchical representations ranging from low-level features to high-level semantic concepts.

2.2.3.2 Fully Convolutional Networks and Dense Prediction

Fully Convolutional Networks (FCNs) replace the fully connected layers of traditional CNNs with convolutional layers, enabling dense, per-pixel predictions without requiring fixed-size inputs. Given an input image of size $H \times W$, an FCN produces a spatial output map of size $H' \times W'$ where each spatial location contains a prediction vector. This makes FCNs naturally suited for dense prediction tasks such as semantic segmentation and object detection.

The absence of fully connected layers preserves spatial relationships and allows variable input sizes. Critically, for nuclei detection, FCN-based architectures are preferred for edge deployment because their computational complexity scales linearly with the number of input pixels, $O(H \times W)$, unlike transformer-based architectures whose self-attention mechanism scales quadratically, $O((H \times W)^2)$. This difference becomes significant for high-resolution histopathology images.

2.2.3.3 Receptive Fields and Large Kernels

The receptive field of a neuron in a CNN is the spatial region in the input that influences its output. For a stack of L convolutional layers each with kernel size k and stride 1, the effective receptive field grows linearly with depth. Traditional architectures use small kernels (typically 3×3) and rely on deep stacking, but this approach suffers from diminishing returns: the effective receptive field grows slower than the theoretical receptive field, and deeper networks become harder to optimize.

Large kernel convolutions (e.g., 31×31) provide an alternative approach to achieving large receptive fields with fewer layers. LKCell [13] demonstrated that transferring pre-trained large kernel models to histopathology significantly improves nuclei segmentation accuracy while reducing computational cost. The key insight is that large kernels achieve broad spatial context in a single layer, capturing tissue architecture and cell

neighborhood information that is critical for accurate boundary prediction. Using depthwise separable convolutions, the computational overhead of large kernels can be kept manageable, making them practical for efficient FCN-based architectures.

2.2.4 Nuclei Detection

2.2.4.1 Detection Paradigms

Object detection frameworks locate and classify objects within images by producing bounding boxes with associated class labels and confidence scores. Two dominant paradigms exist in the literature.

Two-stage detectors, exemplified by Faster R-CNN [5], first propose candidate regions using a Region Proposal Network (RPN) and then classify and refine each proposal independently. This approach achieves high accuracy but is computationally expensive due to the per-region processing.

Single-stage detectors, exemplified by YOLO [6], simultaneously predict bounding boxes and class probabilities across a spatial grid in a single forward pass. This paradigm is significantly faster and more suitable for real-time applications, making it the preferred choice for resource-constrained edge deployment. YOLO divides the input image into an $S \times S$ grid, and each grid cell predicts B bounding boxes with confidence scores and C class probabilities.

Both paradigms output axis-aligned bounding boxes defined by (x_c, y_c, w, h), where (x_c, y_c) is the box center and (w, h) are the width and height. While efficient for localization, this rectangular representation discards all boundary geometry.

2.2.4.2 Non-Maximum Suppression

Non-maximum suppression (NMS) is a post-processing step used by all conventional detection frameworks to eliminate redundant detections. Since detectors typically produce multiple overlapping predictions for a single object, NMS selects the highest-scoring prediction and suppresses all others whose overlap, measured by Intersection over Union (IoU), exceeds a threshold θ_{NMS} (typically 0.5). Formally, given a set of detections sorted by confidence score $\{d_1, d_2, \ldots, d_N\}$, NMS iteratively selects the highest-scoring detection d_i and removes all d_j where:

$$\text{IoU}(d_i, d_j) > \theta_{\text{NMS}} \tag{2-7}$$

While effective for isolated objects, NMS is problematic for nuclei detection because densely packed nuclei produce legitimately overlapping predictions. In clumped regions, two adjacent nuclei may have bounding boxes with high IoU, causing NMS to

suppress the correct detection of one nucleus. This leads to systematic undercounting in precisely the regions where accurate counting matters most for clinical assessment such as sTILs scoring. The severity of this problem increases with nuclear density, creating an inverse relationship between detection accuracy and the clinical importance of the region.

2.2.4.3 Limitations of Detection for Nuclei Analysis

The bounding box representation has two fundamental limitations for nuclei analysis. First, bounding boxes discard boundary geometry. For sTILs scoring, the clinical standard requires measuring the *area* occupied by lymphocytes, not merely counting their centroids. A bounding box provides only a rough rectangular approximation that significantly overestimates cell area, introducing systematic bias into the sTILs calculation. Second, as described above, NMS-based post-processing undercounts in dense regions. Together, these limitations mean that detection models are insufficient for the area-based measurements required in clinical workflows, despite their computational efficiency.

2.2.5 Nuclei Segmentation

2.2.5.1 Instance Segmentation Paradigm

Instance segmentation combines object detection with pixel-level boundary delineation, producing a separate binary mask for each individual object. This addresses the geometry limitation of detection models by providing accurate per-cell boundaries. The dominant approach in general computer vision is Mask R-CNN [3], which extends Faster R-CNN with a mask prediction branch that outputs a binary segmentation mask for each detected region.

In histopathology, U-Net [2] and its variants have been widely adopted. U-Net uses an encoder-decoder architecture with skip connections that preserve fine-grained spatial information. The encoder extracts features at progressively lower resolutions, while the decoder upsamples these features back to the input resolution. Skip connections concatenate encoder features with decoder features, enabling precise boundary localization.

HoVer-Net [7] introduced a specialized approach for nuclei by predicting horizontal and vertical distance maps from each pixel to its nearest nucleus centroid, providing additional discriminative signal for separating touching nuclei during post-processing.

2.2.5.2 Limitations of Segmentation for Nuclei Analysis

Instance segmentation methods face two significant limitations in the context of nuclei analysis.

First, the computational cost is high. Pixel-level mask prediction requires high-resolution feature maps and substantial memory, making these methods impractical for

real-time analysis of whole-slide images on resource-constrained hardware. The post-processing step required to convert dense predictions into individual instances (e.g., watershed, connected component analysis) adds further computational overhead.

Second, the clumped cell problem persists. FCN-based segmentation models produce contiguous probability maps, and when nuclei are tightly packed, the predicted masks for adjacent cells merge into a single connected region. Post-processing methods such as watershed can separate some instances, but they frequently fail in regions of high nuclear density where boundary information is ambiguous. This merging leads to systematic undercounting because merged cells are counted as a single instance. Note that this computational burden stems not from the convolutional architecture itself, but from the costly post-processing required to separate individual instances.

2.2.6 Polygon-Based Instance Segmentation

2.2.6.1 Star-Convex Polygon Representation

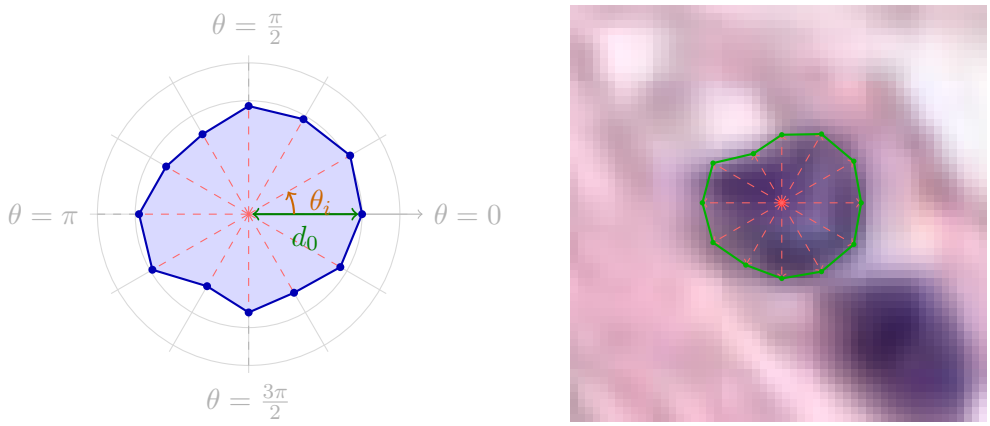

Figure 2.1. Star-convex polygon representation. (a) A nucleus boundary is represented by a centroid $\mathbf{c}$ and n radial distances $\{d_0, d_1, \ldots, d_{n-1}\}$ measured at fixed angular intervals $\theta_i = \frac{2\pi i}{n}$. (b) The representation applied to a real cell image, where the polygon approximates the cell boundary.

Star-convex polygons offer a compact parametric representation of cell boundaries that bridges the gap between detection efficiency and segmentation accuracy. A star-convex polygon is defined by a center point $\mathbf{c} = (c_x, c_y)$ and a set of n radial distances $\{d_1, d_2, \ldots, d_n\}$ measured at fixed angular intervals $\theta_i = \frac{2\pi i}{n}$ for $i = 0, 1, \ldots, n-1$. The polygon boundary is reconstructed as:

$$\mathbf{p}_i = \mathbf{c} + d_i \cdot [\cos\theta_i, \sin\theta_i]^T \tag{2-8}$$

StarDist [8] introduced this representation for cell detection in fluorescence microscopy, later extended to histopathology images [11]. The representation is efficient

because it parameterizes the entire boundary with $n + 2$ values (center coordinates plus n distances), compared to the thousands of pixels required for a segmentation mask. However, StarDist operates within a dense prediction framework followed by NMS-based post-processing, inheriting the undercounting problem in clumped regions.

2.2.6.2 PolarMask: Polar IoU and Centerness

PolarMask [15] integrated the polar polygon representation into a single-shot detection framework, eliminating the need for a separate dense prediction stage. PolarMask formulates instance segmentation as two simultaneous tasks: (1) center point classification, predicting whether each spatial location is an object center, and (2) dense distance regression, predicting n radial distances at each positive center location.

PolarMask introduced two key innovations. The **PolarIoU loss** directly optimizes the overlap between the predicted and ground truth polygons by computing IoU in polar coordinates, providing a more geometrically meaningful supervision signal than individual distance regression losses. The **centerness branch** predicts a scalar value for each spatial location indicating how close it is to the true object center, defined as:

$$\text{centerness} = \sqrt{\frac{\min(l^*, r^*)}{\max(l^*, r^*)} \times \frac{\min(t^*, b^*)}{\max(t^*, b^*)}} \tag{2-9}$$

where l^*, r^*, t^*, b^* are the distances from the location to the left, right, top, and bottom boundaries of the object. This down-weights predictions from locations far from the true center, reducing low-quality detections and improving overall accuracy.

2.2.6.3 Boundary Refinement (CPP-Net)

CPP-Net [12] identified that predicting radial distances from a single centroid pixel provides insufficient contextual information for robust boundary prediction. To address this, CPP-Net samples a *point set* rather than one single pixel within each cell for distance prediction, substantially enhancing the contextual information available to the model.

The predictions from the sampled point set are adaptively fused using a **Confidence-based Weighting Module** that learns to assign higher weights to predictions from more informative sampling locations. Additionally, CPP-Net introduced a **Shape-Aware Perceptual (SAP) loss** that constrains the shape of predicted polygons by comparing them against a reference representation through a pre-trained auxiliary network. Despite these improvements, CPP-Net still operates within a dense prediction framework with NMS-based post-processing, inheriting the same limitations regarding clumped cell separation.

2.2.7 End-to-End Detection with Bipartite Matching

2.2.7.1 Set Prediction and the Hungarian Algorithm

The DETR (DEtection TRansformer) framework introduced a paradigm shift by replacing hand-designed components such as anchor boxes and NMS with a set prediction approach. DETR formulates detection as a bipartite matching problem between a fixed-size set of N predictions $\hat{\mathcal{Y}} = \{\hat{y}_1, \ldots, \hat{y}_N\}$ and ground truth objects $\mathcal{Y} = \{y_1, \ldots, y_M\}$ (with $N \geq M$, padded with $\emptyset$ for the "no object" class).

The optimal assignment σ^* is found by solving:

$$\sigma^* = \arg\min_{\sigma \in \Pi_N} \sum_{i=1}^{N} \mathcal{L}_{\text{match}}(y_i, \hat{y}_{\sigma(i)}) \tag{2-10}$$

where Π_N is the set of all permutations of N elements, and $\mathcal{L}_{\text{match}}$ is a pairwise matching cost that typically combines classification loss, bounding box ℓ_1 loss, and IoU loss. This optimization is solved exactly using the Hungarian algorithm in $O(N^3)$ time. Once the optimal assignment is found, the model is trained end-to-end with a Hungarian loss computed only on matched pairs:

$$\mathcal{L}_{\text{Hungarian}} = \sum_{i=1}^{N} \mathcal{L}(y_i, \hat{y}_{\sigma^*(i)}) \tag{2-11}$$

This formulation ensures that each ground truth object is matched to exactly one prediction, eliminating the need for NMS entirely. The key advantage for nuclei detection is that overlapping predictions are no longer suppressed; instead, the learned assignment naturally resolves which prediction corresponds to which nucleus.

2.2.7.2 LSP-DETR: Star-Convex Polygons with Learned Assignment

LSP-DETR [9] adapted the end-to-end detection framework to nuclei segmentation by combining bipartite matching with star-convex polygon representation. Each prediction consists of a class label, bounding box coordinates, and a set of radial distances defining a star-convex polygon boundary. The matching cost and training loss are extended to include a radial distance term that measures the accuracy of the predicted polygon shape.

The Hungarian matching assigns each ground truth nucleus to exactly one prediction, and the model is trained end-to-end with a combined loss that includes classification, bounding box, and polygon distance terms. This eliminates NMS entirely, resolving the undercounting problem in clumped regions: star-convex polygons naturally overlap in dense areas without being suppressed, enabling accurate instance separation.

LSP-DETR demonstrated strong generalization across tissue types and was over

five times faster than the next-fastest leading method. However, its transformer-based backbone imposes quadratic computational complexity, making it demanding for edge deployment. RayCastED adopts LSP-DETR's assignment mechanism and polygon representation while replacing the transformer backbone with an FCN, preserving the benefits of end-to-end matching with reduced computational cost suitable for resource-constrained environments.

2.3 Analisis Perbandingan Metode

Di dalam tinjauan pustaka hasil akhirnya adalah analisis secara kualitatif atau pun secara kuantitatif kelebihan dan kekurangan metode jika dikaitkan dengan masalah, batasan-batasan masalah dan solusi yang dinginkan. Analisis kuantitatif tidak wajib teapi mempunyai nilai tambah di dalam tugas akhir saudara. Bagian ini menjelaskan kenapa metode tersebut dipilih dan uraikan dengan lebih jelas metode pelaksanaan tugas akhir yang ingin Anda lakukan.

2.4 Pertanyaan Tugas Akhir (Jika Perlu)

Pertanyaan tugas akhir bersifat opsional dan dapat ditambahkan untuk menekankan hal-hal yang hendak diketahui dari tugas akhir berdasar pada tujuan tugas akhir. Pertanyaan tugas akhir dikenal dengan RQ (*Research Question*) dan harus memiliki keterkaitan dengan RO (*Research Objective*). Satu RO dapat memiliki satu atau lebih dari satu RQ.

CHAPTER III
RESEARCH METHODOLOGY

3.1 Datasets

This study uses publicly available secondary datasets for training and evaluation. Table 4.1 summarizes the datasets and their roles.

Table 3.1. Summary of datasets used in this study.

Dataset	Purpose	Tissue	Format
PanNuke	Primary benchmarking	Multi-organ	Pickled bytes
PUMA	TILs detection	Melanoma	TIFF, GeoJSON
panopTILs	TILs detection	Breast	PNG
MoNuSAC	TILs detection	Multi-organ	Pickled bytes

3.1.1 PanNuke

PanNuke [16, 17] is a large-scale pan-cancer nucleus instance segmentation dataset containing approximately 189,744 labeled nuclei across 19 tissue types. Each nucleus is annotated with an instance mask and one of five nuclear categories. The dataset is provided as pickled bytes containing image patches and corresponding label masks. In this study, PanNuke serves as the primary benchmarking dataset for evaluating nuclei detection and segmentation performance.

3.1.2 PUMA

PUMA (Melanoma Histopathology Dataset with Tissue and Nuclei Annotations) [18] provides annotated histopathology images for TILs detection in melanoma tissue. The dataset is distributed as TIFF images paired with GeoJSON annotation files containing polygon-level cell boundary annotations. This dataset is used to evaluate RayCastED's ability to detect and delineate TILs in a clinically relevant setting.

3.1.3 panopTILs

panopTILs [19] provides TILs annotations on breast cancer histopathology images. The dataset is distributed as PNG image patches with corresponding annotation files. This dataset supplements the TILs evaluation alongside PUMA, providing breast tissue samples for assessing TILs detection performance across cancer types.

3.1.4 MoNuSAC

MoNuSAC [20] provides annotated nuclei across multiple organ types with four nuclear categories: epithelial, lymphocyte, neutrophil, and macrophage. The dataset is provided as pickled bytes. The lymphocyte annotations in MoNuSAC are particularly relevant for evaluating TILs detection, as lymphocytes are the primary cell type of interest in TILs scoring.

3.2 Methods

Bagian ini membahas metode atau cara yang akan digunakan dalam penelitian, tahapan penerapan metode, dan desain penelitian (misalnya apakah penelitian akan menggunakan eksperimen di Laboratorium atau di lapangan, misalkan saja penelitian biomedis atau penelitian alat ukur hama yang dapat dilakukan di laboratorium ataupun di lapangan, atau menggunakan metode survei (misalnya untuk teknologi Informasi), studi kasus, atau analisis dengan perangkat lunak (ETAP, LTSpice, dst), atau *prototyping* (pembuatan perangkat keras).

Bagian ini juga membahas bagaimana data [akan] dianalisis, apakah dengan membandingkan keluaran beberapa alat ukur, membandingkan dengan standar atau bagaimana.

3.3 Research Workflow

Menguraikan prosedur yang akan digunakan dan jadwal atau alur penyelesaian setiap tahap. Alur penelian ini dapat disajikan dalam bentuk diagram. Diagram dapat disusun dengan aturan yang baik semisal menggunakan *flowchart*. Aturan dan tutorial pembuatan *flowchart* dapat dilihat di http://ugm.id/flowcharttutorial. Setelah menggambarkannya, penulis wajib menjelaskan langkah-langkah setiap alur tugas akhir dalam sub bab tersendiri sesuai dengan kebutuhan.

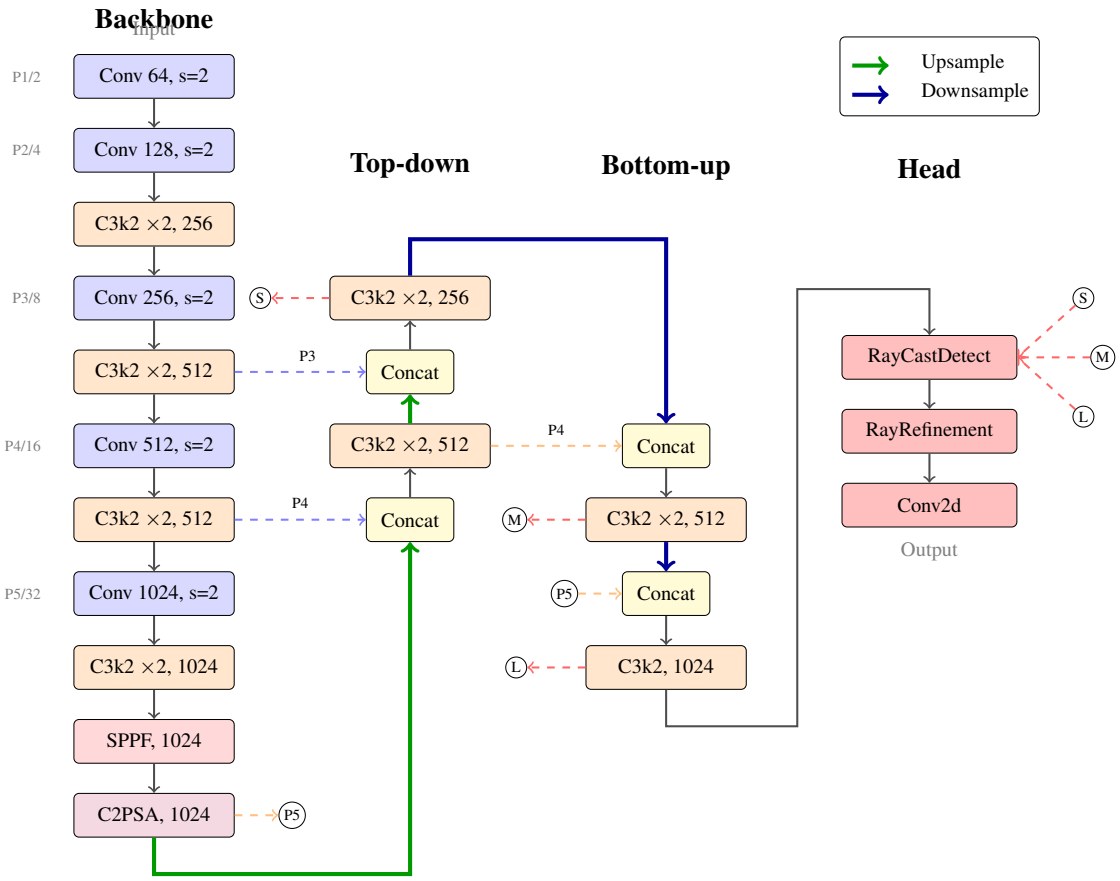

Figure 3.2. YOLO26 architecture.

CHAPTER IV
HASIL DAN PEMBAHASAN

Table 4.1. Evaluation Against LSP-DETR

Metric	LSP-DETR	RayCastED	delta
AJI	**0.825**	0.6385	-0.1865
FI	**0.677**	0.5581	-0.1189
mAP@0.5	**0.691**	0.4275	-0.2635
Precision	**0.862**	0.6848	-0.1772
Recall	**0.791**	0.5980	-0.1930
PQ	**0.657**	0.5433	-0.1137
SQ	**0.811**	0.7460	-0.0650
DQ	**0.810**	0.6817	-0.1283
Params	45.0M	**11.92M**	-33.08M
GFLOPs	-	4.77	-
Inference Time	-	4.17ms	-

Berikut ini adalah yang perlu diperhatikan untuk mengisi bab hasil dan pembahasan:

1. Setiap rumusan masalah boleh memiliki lebih dari 1 tujuan.
2. Setiap subbab harus spesifik menjawab setiap tujuan yang dituliskan.
3. Setiap rumusan masalah boleh dijawab dengan 1 subbab atau lebih.

Berikut ini adalah contoh sub bab untuk menjelaskan tujuan penelitian.

4.1 Pembahasan Tujuan 1 dengan Hasil Penelitian 1 (Ubah Judul Sesuai dengan Hal yang Hendak dibahas)

Sub bab pertama adalah membahas tujuan penelitian pertama dengan hasil penelitian ke-1. Dapat ditambahkan beberapa sub bab jika diperlukan.

4.2 Pembahasan Tujuan 1 dengan Hasil Penelitian 2 (Ubah Judul Sesuai dengan Hal yang Hendak dibahas)

Sub bab kedua adalah membahas tujuan penelitian pertama dengan hasil penelitian ke-2. Sub bab ini merupakan contoh tambahan sub bab pertama.

4.3 Pembahasan Tujuan 2 dengan Hasil Penelitian 3 (Ubah Judul Sesuai dengan Hal yang Hendak dibahas)

Sub bab ketiga adalah membahas tujuan penelitian kedua. Dapat ditambahkan beberapa sub bab jika diperlukan.

4.4 Perbandingan Hasil Penelitian dengan Hasil Terdahulu

Pembahasan penutup dapat menjelaskan mengenai kelebihan hasil pengembangan / penelitian dan kekurangan dibandingkan dengan skripsi atau penelitian terdahulu atau perbandingan terhadap produk lain yang ada di pasaran. Penulis dapat menggunakan tabel untuk membandingkan secara gamblang dan menjelaskannya.

CHAPTER V
TAMBAHAN (OPSIONAL)

Anda boleh menambahkan Bab jika diperlukan. Jumlah Bab tidak harus sesuai dengan *template*.

Bab tambahan ini diperlukan jika hasil penelitian untuk menjawab tujuan cukup panjang atau terdiri dari banyak sub bab. Mahasiswa boleh menjawab 1 tujuan penelitian dengan 1 bab.

CHAPTER VI
KESIMPULAN DAN SARAN

6.1 Kesimpulan

Kesimpulan dapat diawali dengan apa yang dilakukan dengan tugas akhir ini lalu dilanjutkan dengan poin-poin yang menjawab tujuan penelitian, apakah tujuan sudah tercapai atau belum, tentunya berdasarkan data ataupun hasil dari Bab pembahasan sebelumnya. Dalam beberapa hal, kesimpulan dapat juga berisi tentang temuan/*findings* yang Anda dapatkan setelah melakukan pengamatan dan atau analisis terhadap hasil penelitian.

Kesimpulan menjawab seberapa jauh rumusan masalah tercapai berdasarkan hasil penelitian. Semua rumusan masalah harus disimpulkan berdasarkan data penelitian.

6.2 Saran

Saran berisi hal-hal yang bisa dilanjutkan dari penelitian atau skripsi ini, yang belum dilakukan karena batasan permasalahan. Saran bukan berisi saran kepada sistem atau pengguna, tetapi saran diberikan kepada aspek penelitian yang dapat dikembangkan dan ditambahkan di penelitian atau skripsi selanjutnya.

Catatan: Mahasiswa perlu melihat sinkronisasi antara rumusan masalah, tujuan, metode, hasil penelitian, dan kesimpulan.

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Catatan: Daftar pustaka adalah apa yang dirujuk atau disitasi, bukan apa yang telah dibaca, jika tidak ada dalam sitasi maka tidak perlu dituliskan dalam daftar pustaka.

LAMPIRAN

L.1 Isi Lampiran

Lampiran bersifat opsional bergantung hasil kesepakatan dengan pembimbing dapat berupa:

1. Bukti pelaksanaan Kuesioner seperti pertanyaan kuesioner, resume jawaban responden, dan dokumentasi kuesioner.
2. Spesifikasi Aplikasi atau Sistem yang dikembangkan meliputi spesifikasi teknis aplikasi, tautan unduh aplikasi, manual penggunaan aplikasi, hingga screenshot aplikasi.
3. Cuplikan kode yang sekiranya penting dan ditambahkan.
4. Tabel yang terlalu panjang yang masih diperlukan tetapi tidak memungkinkan untuk ditayangkan di bagian utama skripsi.
5. Gambar-gambar pendukung yang tidak terlalu penting untuk ditampilkan di bagian utama. Akan tetapi, mendukung argumentasi/pengamatan/analisis.
6. Penurunan rumus-rumus atau pembuktian suatu teorema yang terlalu panjang dan terlalu teknis sehingga Anda berasumsi bahwa pembaca biasa tidak akan menelaah lebih lanjut. Hal ini digunakan untuk memberikan kesempatan bagi pembaca tingkat lanjut untuk melihat proses penurunan rumus-rumus ini.

LAMPIRAN

L.2 Panduan Latex

L.2.1 Syntax Dasar

L.2.1.1 Penggunaan Sitasi

Contoh penggunaan sitasi [21, 22] [23] [24] [25] [26, 27]

L.2.1.2 Penulisan Gambar

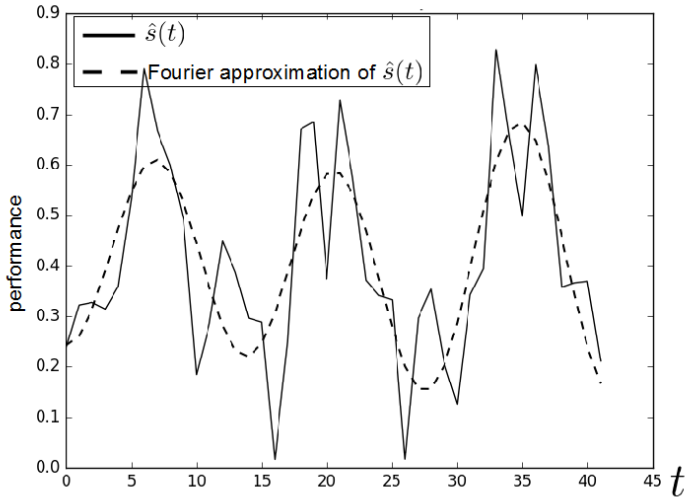

Figure L.1. Contoh gambar.

Contoh gambar terlihat pada Gambar L.1. Gambar diambil dari [27].

L.2.1.3 Penulisan Tabel

Table L.1. Tabel ini

ID	Tinggi Badan (cm)	Berat Badan (kg)
A23	173	62
A25	185	78
A10	162	70

Contoh penulisan tabel bisa dilihat pada Tabel L.1.

L.2.1.4 Penulisan formula

Contoh penulisan formula

$$L_{\psi_z} = \{t_i \mid v_z(t_i) \leq \psi_z\} \tag{1}$$

Contoh penulisan secara *inline*: $PV = nRT$. Untuk kasus-kasus tertentu, kita membutuhkan perintah "mathit" dalam penulisan formula untuk menghindari adanya jeda saat penulisan formula.

Contoh formula **tanpa** menggunakan "mathit": $PVA = RTD$

Contoh formula **dengan** menggunakan "mathit": $PVA = RTD$

L.2.1.5 Contoh list

Berikut contoh penggunaan list

1. First item
2. Second item
3. Third item

L.2.2 Blok Beda Halaman

L.2.2.1 Membuat algoritma terpisah

Untuk membuat algoritma terpisah seperti pada contoh berikut, kita dapat memanfaatkan perintah *algstore* dan *algrestore* yang terdapat pada paket *algcompatible*. Pada dasarnya, kita membuat dua blok algoritma dimana blok pertama kita simpan menggunakan *algstore* dan kemudian di-restore menggunakan *algrestore* pada algoritma kedua. Perintah tersebut dimaksudkan agar terdapat kesinamungan antara kedua blok yang sejatinya adalah satu blok.

Algorithm 1 Contoh algorima

```
procedure CREATESET(v)
    Create new set containing v
end procedure
```

Pada blok algoritma kedua, tidak perlu ditambahkan caption dan label, karena sudah menjadi satu bagian dalam blok pertama. Pembagian algoritma menjadi dua bagian ini berguna jika kita ingin menjelaskan bagian-bagian dari sebuah algoritma, maupun untuk memisah algoritma panjang dalam beberapa halaman.

```
procedure CONCATSET(v)
    Create new set containing v
end procedure
```

L.2.2.2 Membuat tabel terpisah

Untuk membuat tabel panjang yang melebihi satu halaman, kita dapat mengganti kombinasi *table* + *tabular* menjadi *longtable* dengan contoh sebagai berikut.

Table L.2. Contoh tabel panjang

header 1	header 2
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar

L.2.2.3 Menulis formula terpisah halaman

Terkadang kita butuh untuk menuliskan rangkaian formula dalam jumlah besar sehingga melewati batas satu halaman. Solusi yang digunakan bisa saja dengan memindahkan satu blok formula tersebut pada halaman yang baru atau memisah rangkaian formula menjadi dua bagian untuk masing-masing halaman. Cara yang pertama mungkin akan menghasilkan alur yang berbeda karena ruang kosong pada halaman pertama akan diisi oleh teks selanjutnya. Sehingga di sini kita dapat memanfaatkan *align* yang sudah diatur dengan mode *allowdisplaybreaks*. Penggunakan *align* ini memungkinkan satu rangkaian formula terpisah berbeda halaman.

Contoh sederhana dapat digambarkan sebagai berikut.

$$\begin{aligned} x &= y^2 \\ x &= y^3 \\ a + b &= c \\ x &= y - 2 \\ a + b &= d + e \\ x^2 + 3 &= y \\ a(x) &= 2x \end{aligned} \tag{2}$$

$$
\begin{aligned}
b_i &= 5x \\
10x^2 &= 9x \\
2x^2 + 3x + 2 &= 0 \\
5x - 2 &= 0 \\
d &= \log x \\
y &= \sin x
\end{aligned}
$$

LAMPIRAN

L.3 Format Penulisan Referensi

Penulisan referensi mengikuti aturan standar yang sudah ditentukan. Untuk internasionalisasi DTETI, maka penulisan referensi akan mengikuti standar yang ditetapkan oleh IEEE (*International Electronics and Electrical Engineers*). Aturan penulisan ini bisa diunduh di http://www.ieee.org/documents/ieeecitationref.pdf. Gunakan Mendeley sebagai *reference manager* dan *export* data ke format Bibtex untuk digunakan di Latex.

Berikut ini adalah sampel penulisan dalam format IEEE:

L.3.1 Book

Basic Format:

[1] J. K. Author, "Title of chapter in the book," in Title of His Published Book, xth ed. City of Publisher, Country: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx–xxx.

Examples:

[1] B. Klaus and P. Horn, Robot Vision. Cambridge, MA: MIT Press, 1986.

[2] L. Stein, "Random patterns," in Computers and You, J. S. Brake, Ed. New York: Wiley, 1994, pp. 55-70.

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Polymers of Hexadromicon, J. Peters, Ed., 2nd ed. New York: McGraw-Hill, 1964, pp. 15-64.

L.3.2 Handbook

Basic Format:

[1] Name of Manual/Handbook, x ed., Abbrev. Name of Co., City of Co., Abbrev. State, year, pp. xx-xx.

Examples:

[1] Transmission Systems for Communications, 3rd ed., Western Electric Co., Winston Salem, NC, 1985, pp. 44-60.

[2] Motorola Semiconductor Data Manual, Motorola Semiconductor Products Inc., Phoenix, AZ, 1989.

[3] RCA Receiving Tube Manual, Radio Corp. of America, Electronic Components and Devices, Harrison, NJ, Tech. Ser. RC-23, 1992.

Conference/Prosiding

Basic Format:

[1] J. K. Author, "Title of paper," in Unabbreviated Name of Conf., City of Conf., Abbrev. State (if given), year, pp.xxx-xxx.

Examples:

[1] J. K. Author [two authors: J. K. Author and A. N. Writer] [three or more authors: J. K. Author et al.], "Title of Article," in [Title of Conf. Record as], [copyright year] © [IEEE or applicable copyright holder of the Conference Record]. doi: [DOI number]

Sumber Online/Internet

Basic Format:

[1] J. K. Author. (year, month day). Title (edition) [Type of medium]. Available: http://www.(URL)

Examples:

[1] J. Jones. (1991, May 10). Networks (2nd ed.) [Online]. Available: http://www.atm.com

Skripsi, Tesis dan Disertasi

Basic Format:

[1] J. K. Author, "Title of thesis," M.S. thesis, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

[2] J. K. Author, “Title of dissertation,” Ph.D. dissertation, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

Examples:

[1] J. O. Williams, “Narrow-band analyzer,” Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993. [2] N. Kawasaki, “Parametric study of thermal and chemical nonequilibrium nozzle flow,” M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993

L.4 Contoh Source Code

L.4.1 Sample algorithm

Algorithm 2 Kruskal's Algorithm

```
procedure MAKESET(v)
    Create new set containing v
end procedure

function FINDSET(v)
    return a set containing v
end function

procedure UNION(u,v)
    Unites the set that contain u and v into a new set
end procedure

function KRUSKAL(V, E, w)
    A ← {}
    for each vertex v in V do
        MakeSet(v)
    end for
    Arrange E in increasing costs, ordered by w
    for each (u,v) taken from the sorted list do
        if FindSet(u) ≠ FindSet(v) then
            A ← A ∪ {(u, v)}
            Union(u, v)
        end if
    end for
    return A
end function
```

L.4.2 Sample Python code

```
import numpy as np

def incmatrix(genl1,genl2):
  m = len(genl1)
  n = len(genl2)
  M = None #to become the incidence matrix
  VT = np.zeros((n*m,1), int)  #dummy variable

  #compute the bitwise xor matrix
  M1 = bitxormatrix(genl1)
  M2 = np.triu(bitxormatrix(genl2),1)

  for i in range(m-1):
    for j in range(i+1, m):
      [r,c] = np.where(M2 == M1[i,j])
      for k in range(len(r)):
        VT[(i)*n + r[k]] = 1;
        VT[(i)*n + c[k]] = 1;
        VT[(j)*n + r[k]] = 1;
        VT[(j)*n + c[k]] = 1;

  if M is None:
    M = np.copy(VT)
  else:
    M = np.concatenate((M, VT), 1)

  VT = np.zeros((n*m,1), int)

  return M
```

L.4.3 Sample Matlab code

```
function X = BitXorMatrix(A,B)
%function to compute the sum without charge of two vectors

  %convert elements into usigned integers
  A = uint8(A);
  B = uint8(B);

  m1 = length(A);
  m2 = length(B);
  X = uint8(zeros(m1, m2));
  for n1=1:m1
    for n2=1:m2
      X(n1, n2) = bitxor(A(n1), B(n2));
    end
  end
```